Lecture 3 Live Problems

State variables and their change equations

Points toward lab manual pages: **BT1, BT2, Pop1, Pop2, SL1, SL2** (section L1 S2)

1. Synchronized cell division. Let t be time in days after fertilization, and let C be the number of cells arising from the fertilized egg. Fill in the table, and compute the rate of growth in each interval. How is ΔC related to C itself?

t (days)	Δt	C (cells)	ΔC	$\Delta C / \Delta t$
0		1		
1		2		
2		4		
3		8		
4		16		
5		32		

2. Relative change. Suppose we have 200 cells.

(a) If the population increases by 10%, how many cells?

(b) If the population decreases by 10%, how many cells?

(c) If the population increases by 100%, how many cells?

(d) If the population increases by 50%, and then increases by 50% again, how many cells?

3. Caffeine. Let C be the amount of caffeine in my body, in milligrams, and let t be time in hours. Every hour, about 20% of the caffeine in my body is metabolized, so $\Delta C/\Delta t = -0.2\,C$. Suppose I have 250 mg of caffeine in my body at 3 pm, and I take in no more caffeine. Tabulate and plot the expected caffeine in my body from 3 pm until 8 pm.

t (hours after 3 pm)	Δt	C (mg)	ΔC
0		250	
1			
2			
3			
4			
5			

